

When Does Fourier Diagonalize the Physics?

Calibrated Meters for Fixed versus Learned Representations
in Operator Learning, and What the Learned Part Is Worth

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Abstract — A companion study showed that, for *classification*, a pooled dictionary of fixed transforms usually beats a transform learned from scratch, and that a gated coupling of the two self-calibrates: the gate closes where the dictionary suffices and opens where it does not [10]. This paper re-instantiates that architecture for *physical modelling*, where the objective inverts — predict the state rather than compress toward a label — and asks a question that classification cannot answer well: *what does the learned part actually say?* We report six results, all on controlled one-dimensional partial differential equations (PDEs). (i) The failure of a fixed spectral basis to *diagonalize* the propagator is a clean, monotone **heterogeneity meter**: for diffusion with a spatially varying coefficient $a(x)$, Fourier’s off-diagonal fraction rises $0 \rightarrow 0.273$ with the heterogeneity knob while the true eigenbasis stays exactly diagonal. (ii) That eigenbasis can be *discovered without an oracle* by minimising off-diagonality over a *structured* Sturm–Liouville family — reaching 0.000000 at every heterogeneity and returning the physical coefficient at correlation 1.0000 — and the same objective detects its own mis-specification. (iii) The residual of a data-driven linear (dynamic mode decomposition, DMD) propagator is a **nonlinearity meter**, and a gated learned branch fires with the *same rectified, thresholded law* as the classification gate: shut until the nonlinearity knob $\varepsilon \approx 0.4$, then rising linearly. (iv) Making that branch *deeper* is not the lever; making it *structurally correct* is — held-out capture of what DMD misses goes $11.4\% \rightarrow 71.6\%$ when the branch’s features match the true quadratic term, while stacked residual integrators reach only 15.6%. (v) Sparse regression then *names* that term exactly, recovering the diffusion coefficient to three decimals and the nonlinear coefficient on the identity line. (vi) Putting the named term into the model buys **extrapolation**: over a 30-step rollout never seen in training it is $\sim 4\times$ more accurate than a black-box branch, and $\sim 3\times$ on unseen amplitudes; rollout training then improves the structured model a further $3.4\times$ while making the black box *worse*. We are explicit about what is established elsewhere (operator learning, Koopman/DMD, sparse identification, inverse coefficient problems) and about our own negative results, discarded metrics and confound controls.

1 Introduction

Two families of front-end compete whenever a signal must be represented compactly. A *fixed* transform — Fourier, wavelet, scattering — is cheap, interpretable and needs no data. A *learned* transform is adaptable but under-determined and opaque. A companion study [10] compared them for classification and found that a gated coupling behaves best, with the gate acting as a diagnostic: it collapses to the dictionary when the dictionary suffices, and engages the learned branch only where the discriminative structure is genuinely non-classical.

That study left a dissatisfaction. Under a classification objective, *reading* the learned transform is ill-posed: many bases separate the same labels, so the learned basis drifts and its entries carry

	classification (companion)	physical modelling (this paper)
objective	compress toward the label	predict/reconstruct the state
fixed branch	Fourier/wavelet as features	spectral method: Fourier <i>diagonalizes</i> homogeneous c
learned branch	discriminative basis (under-determined)	eigenbasis of the dynamics (determined by the physic
head	classifier	propagator $u(t) \rightarrow u(t+\Delta)$
gate	is the discriminative basis outside the bank?	<i>is the physics diagonalized by a classical transform?</i>
reading the basis	ill-posed (drifts)	<i>recovering the operator</i> — well-posed

Table 1: The inversion. Every row changes meaning when the objective changes from label compression to state prediction.

no meaning beyond the task. The natural fix is to change the objective. In physical modelling the target is the state itself, and the learned object stops being an arbitrary discriminative rotation: it becomes the basis that *diagonalizes the dynamics*. Table 1 states the inversion.

What we claim, and what we do not. Several ingredients here are established and we use them as tools, not contributions: neural operator learning and its spectral variants [1]; Koopman operator theory and dynamic mode decomposition [2, 3]; sparse identification of governing equations [4, 5]; the recovery of a coefficient from spectral data, which is a mature inverse problem [6]; and the general finding in scientific machine learning that inductive bias beats raw capacity. Relatedly, it has recently been *proved* that an invariant network’s weights recover the Fourier transform of the underlying group, and that the group can be read off those weights [7]; we do not re-derive that. Our contribution is narrower and, we believe, useful: a set of **calibrated meters** — quantities you compute on your own operator or data that answer *how much* of a fixed representation is failing, *of what kind*, and *whether the learned part is worth reading* — plus the controls that make those meters trustworthy.

Roadmap. Section 2 fixes notation and the diagonality metric. Sections 3 and 4 build and then close the heterogeneity meter. Sections 5 and 6 build the nonlinearity meter and settle depth-versus-structure. Section 7 reads the operator, Sections 8 and 9 measure what that reading is worth. Section 11 collects the negative results.

2 Setup

Operators. We use two controlled families on a periodic grid of N points. The *linear* family is diffusion with a spatially varying coefficient, $u_t = \partial_x(a(x) \partial_x u)$, discretized as $L = -G^\top \text{diag}(a_f) G$ with G the forward-difference operator and a_f the coefficient on cell faces; the one-step propagator is $P = \exp(L \Delta t)$. A knob β sets the heterogeneity, $a = 1 + \beta f(x)$ with f a fixed smooth normalized field; $\beta = 0$ gives constant coefficients, for which Fourier is exactly the eigenbasis. The *nonlinear* family is $u_t = \nu u_{xx} - \varepsilon u u_x$, integrated with fourth-order Runge–Kutta and 2/3 dealiasing: $\varepsilon = 0$ is the linear heat equation, $\varepsilon > 0$ is Burgers.

The diagonality metric. For a basis B (rows orthonormal) and a propagator P , define

$$\mathcal{D}(B, P) = \frac{\|BPB^\top - \text{diag}(BPB^\top)\|_F}{\|BPB^\top\|_F}, \quad (1)$$

the off-diagonal Frobenius fraction. $\mathcal{D} = 0$ means a *diagonal* (per-mode) propagator in B reproduces the dynamics exactly — a spectral method is exact and there is nothing to learn.

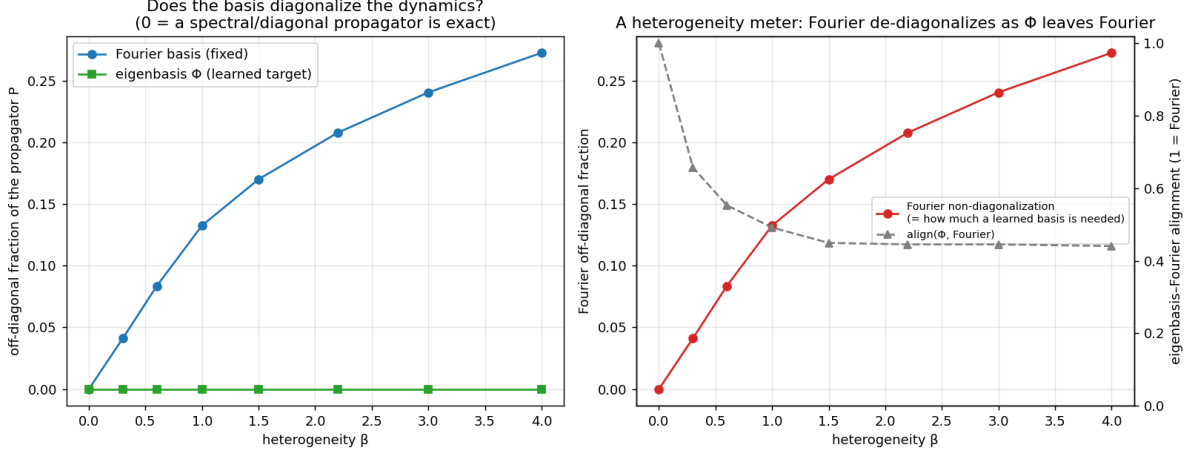


Figure 1: Heterogeneity meter. Left: off-diagonal fraction of the propagator in the fixed Fourier basis (blue) and in the true eigenbasis (green) as the medium becomes heterogeneous. Right: the same Fourier curve as a meter, against the alignment between the true eigenbasis and Fourier.

This is a property of the *operator*, not of sampled states, which matters: an earlier version of this experiment used a prediction error instead and produced an artefact, because initial conditions that are smooth in Fourier let Fourier compress the *data* regardless of the dynamics (Section 11).

3 A heterogeneity meter

We sweep β and evaluate Equation (1) for the fixed real Fourier basis and for the true eigenbasis Φ of L (an oracle, removed in Section 4).

At $\beta = 0$, Fourier diagonalizes P exactly ($\mathcal{D} = 0.000$). As heterogeneity grows, $\mathcal{D}_{\text{Fourier}}$ rises *monotonically* to 0.273 at $\beta = 4$, while $\mathcal{D}_{\Phi} \equiv 0$ at every β ; the alignment between Φ and Fourier falls from 1.00 to 0.44 (Figure 1). The fixed basis is therefore optimal exactly where the medium is homogeneous, and degrades in proportion to how far the physics has moved away from it. *The failure of the fixed basis is the measurement.*

4 Discovering the basis without an oracle

The oracle is the weak point of Section 3. Learning a *free* unitary is the classification failure mode — under-determined and drifting — so instead we search a *structured* family: Sturm–Liouville operators $L(\theta) = -G^{\top} \text{diag}(a_{\theta})G$ with a_{θ} a truncated Fourier parametrization of the coefficient (6 or 10 parameters). The objective is *only* Equation (1) on the true propagator: no labels, no trajectories, no oracle, started from Fourier ($\theta = 0$).

The family is well motivated rather than arbitrary: the classical orthogonal bases (Fourier, Hermite, Legendre, Chebyshev) *are* the eigenbases of Sturm–Liouville operators, so searching over such bases is searching over second-order differential operators. Note also a degeneracy that makes the operator, not the basis, the right thing to parametrize: at a fixed rank K , all polynomial families of degree $< K$ span the *same* subspace.

Result. Off-diagonality falls from 0.041–0.273 (Fourier) to **0.000000** at every β , and the recovered coefficient matches the truth at correlation **1.0000** (Figure 2). Eigenvectors are invariant under $a \rightarrow ca$ for $c > 0$, so a is identifiable only up to a positive scale and recovery is

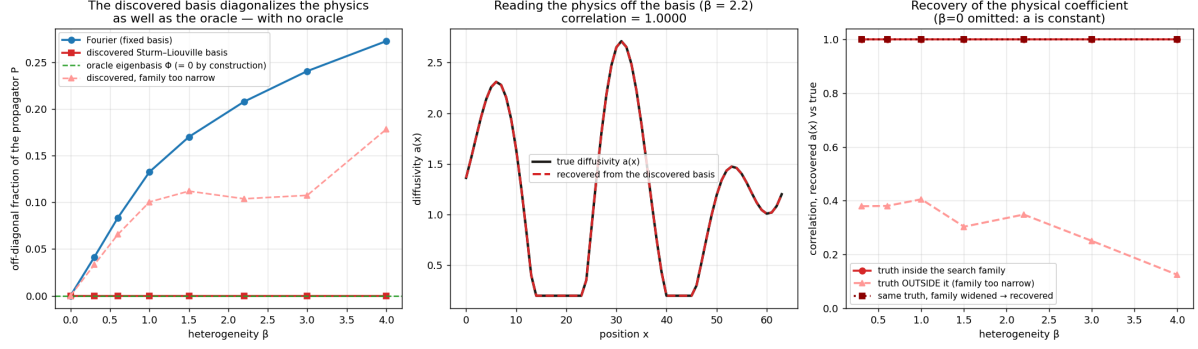


Figure 2: Discovering the basis. Left: the discovered Sturm–Liouville basis diagonalizes as well as the oracle, with no oracle; the pale curve is the same search with a family too narrow to contain the truth. Centre: the recovered coefficient over the true one. Right: recovery against heterogeneity for well-specified, mis-specified and re-specified families.

scored after standardizing both profiles. The discovered basis is not merely a basis that works — it *is* the physics, readable as a coefficient field.

The meter detects its own mis-specification. When the true $a(x)$ contains harmonics *outside* the search family, recovery degrades sharply: off-diagonality only falls $0.263 \rightarrow 0.178$ and the recovered profile correlates $0.13\text{--}0.41$ with the truth. Widening the family to contain the truth restores 0.000000 and 1.0000 . So the failure mode is mis-specification, not the method — and the objective value itself announces it, by plateauing well above zero. This is the precise counterweight to Section 6: structure is the lever, but *wrong* structure is worse than useless.

5 A nonlinearity meter, and a gate that fires

We now move to $u_t = \nu u_{xx} - \varepsilon u u_x$ and learn a one-step map over a horizon long enough for the nonlinearity to act. The linear reference is the closed-form best linear propagator A (DMD): it can only represent a superposition of exponential modes, so its held-out residual is exactly what a Laplace/Koopman-linear front-end cannot capture. On top we place a gated learned branch, $\hat{u}_1 = Au_0 + g \cdot \text{NL}(u_0)$, with A frozen and only the branch trained on the residual under an L_1 penalty on g .

Two things track ε monotonically (Figure 3). The DMD residual rises $0 \rightarrow 0.153$: a clean nonlinearity meter. And the branch *fires with a rectified, thresholded law* — amplitude ≈ 0 up to $\varepsilon \approx 0.4$, then rising roughly linearly to 0.127 . That is the same shape as the classification gate law of the companion study, where the gate followed a controlled information gap with Pearson $r = 0.96$ and a rectified fit. The branch does not fire until the physics is nonlinear enough to need it, and then fires in proportion. When it fires it genuinely captures out-of-sample, removing a steady $11.6\text{--}13.4\%$ of the DMD residual.

6 Depth is not the lever; structure is

The branch above captures only $\sim 13\%$ of what DMD misses. Since $u u_x$ is a low-order polynomial nonlinearity that a two-layer network can express, the ceiling is one of *generalization*, not capacity — confirmed by the fact that removing regularization makes the branch fire more while going *negative* on held-out data. We therefore compare four branch designs on the identical frozen-DMD residual (Table 2, Figure 4).

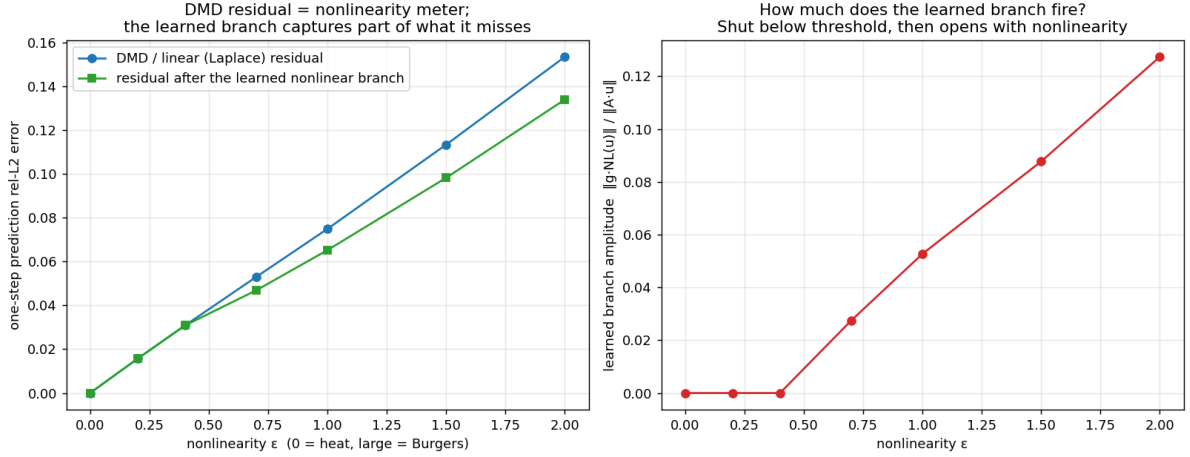


Figure 3: Nonlinearity meter and gate. Left: DMD (linear) residual against the nonlinearity knob, and the residual after the gated learned branch. Right: how much the branch fires — shut below threshold, then opening in proportion.

nonlinear branch	mean held-out capture ($\epsilon \geq 0.7$)
one-step multi-layer perceptron (no depth)	11.4%
residual integrator, 4 stages (neural-ODE-like)	15.6%
residual integrator, 8 stages*	6.1%
structure-matched quadratic $u \odot \partial_x u$	71.6%

Table 2: Depth versus structure. *under-trained at a fixed budget. Adding depth as a learned integrator helps modestly and non-monotonically; matching the form of the nonlinearity multiplies capture by six.

The winning branch is not just better — it *is* the feature that sparse regression selects (Section 7). Capturing the nonlinearity a linear front-end misses and writing down the governing term are therefore the same operation, which is what makes the rest of this paper possible.

7 Reading the operator

We estimate u_t *numerically* (a small forward difference, not the analytic right-hand side), build a library of candidate spatial-differential terms $[u, u_x, u_{xx}, u_{xxx}, u^2, uu_x, uu_{xx}, (u_x)^2]$ via spectral derivatives, and run sequential thresholded least squares.

The recovery is exact in the accessible regime (Figure 5): the diffusion coefficient is pinned at $0.050 = \nu$, the coefficient of uu_x lies on the identity line $-\epsilon$ (measured $-0.200, -0.400, -0.699, -0.991$ for $\epsilon = 0.2, 0.4, 0.7, 1.0$), and every non-physical library term is driven to zero by the sparsity. Honest limit: for $\epsilon \geq 1.5$ the nonlinear coefficient is under-estimated (-1.688 against -2.0) and ν drifts to 0.042 . This is a resolution effect — strong Burgers shocks are under-resolved at $N = 64$ and the numerical u_t degrades where gradients steepen — not a failure of identification.

8 What the reading is worth: extrapolation

Recovering an equation is only valuable if it buys something a black box cannot give. We test the sharpest version: three one-step models sharing the *same frozen* linear DMD head, differing only in the nonlinear branch — (a) none, (b) a black-box network, (c) the discovered uu_x term

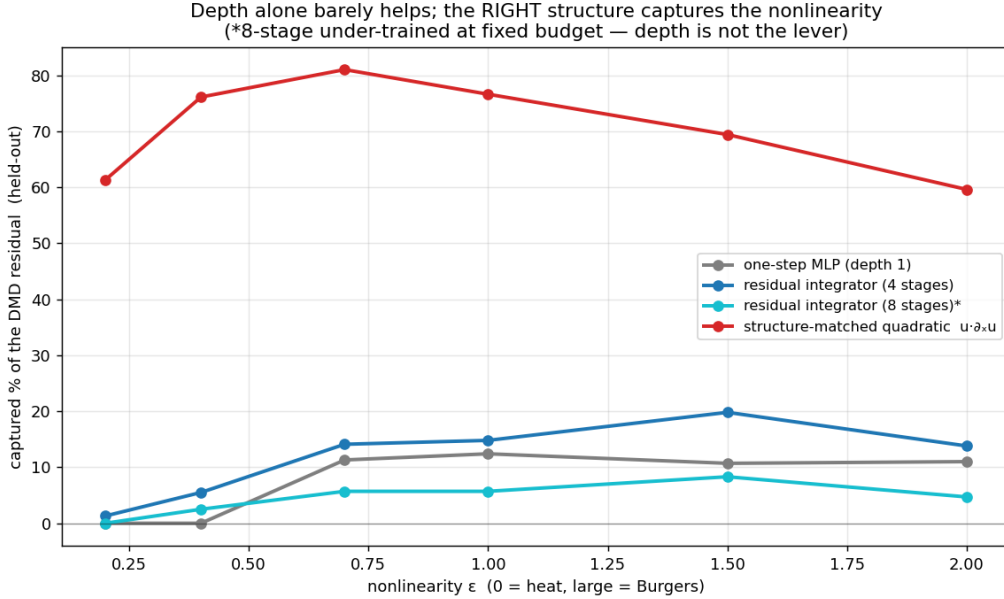


Figure 4: Held-out capture of the DMD residual for four branch designs across the nonlinearity sweep.

— all trained *only on one-step pairs* at $\varepsilon = 1.0$, then evaluated by a 30-step *rollout* (a horizon never seen) from in-distribution and from higher-amplitude out-of-distribution initial conditions.

model	rollout error, in-distribution	rollout error, unseen amplitude
DMD linear only	0.563	0.748
DMD + black-box branch	0.558	0.739
DMD + discovered uu_x	0.150	0.303

Table 3: Relative L_2 error at rollout step 30. The discovered term is $\sim 4\times$ better in horizon and $\sim 2.5\times$ on unseen amplitude.

The most telling line is the middle one: the black-box branch sits essentially *on top of* the linear-only curve at every step (Figure 6). Its $\sim 11\%$ one-step capture does not compound into any rollout advantage — it did not learn a generalizable operator, only a one-step correction on the training distribution. It does not explode; it simply fails to help. The structured term transfers because uu_x is amplitude-covariant: the physically correct term carries to a regime it never saw.

9 Hardening: rollout training multiplies structure

Finally we ask whether training *on* rollout (rather than one-step) changes the picture, and whether the models respect the physics — viscous Burgers must dissipate energy, never create it. Rollout training is applied as fine-tuning on top of the converged one-step model, unrolling 5 steps with gradient clipping; three seeds, mean $\pm$ standard deviation.

Rollout training is therefore *not a substitute for structure; it is a multiplier on structure* (Figure 7). It also collapses the seed variance of the structured model (out-of-distribution standard deviation $0.075 \rightarrow 0.007$): the right operator trained on the right objective becomes seed-insensitive. The one-step structured model reproduces Section 8 within seed variation (0.138 ± 0.012 here against 0.150 there), which is precisely why this table carries error bars. The physical check passes for all models — spurious energy growth is 0.0% of steps everywhere except the black box under

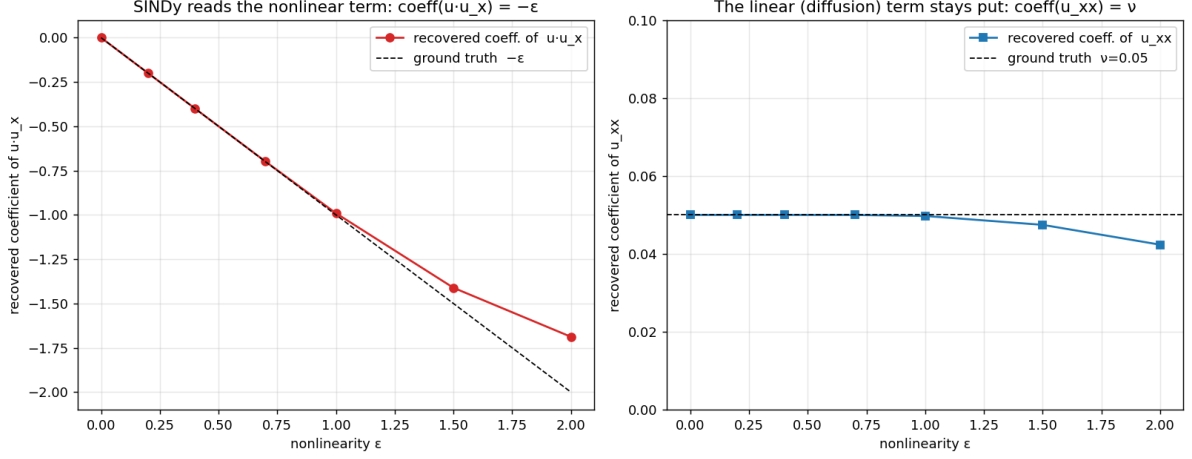


Figure 5: Reading the operator. Left: recovered coefficient of uu_x against ground truth $-\varepsilon$. Right: the linear (diffusion) coefficient stays pinned at ν .

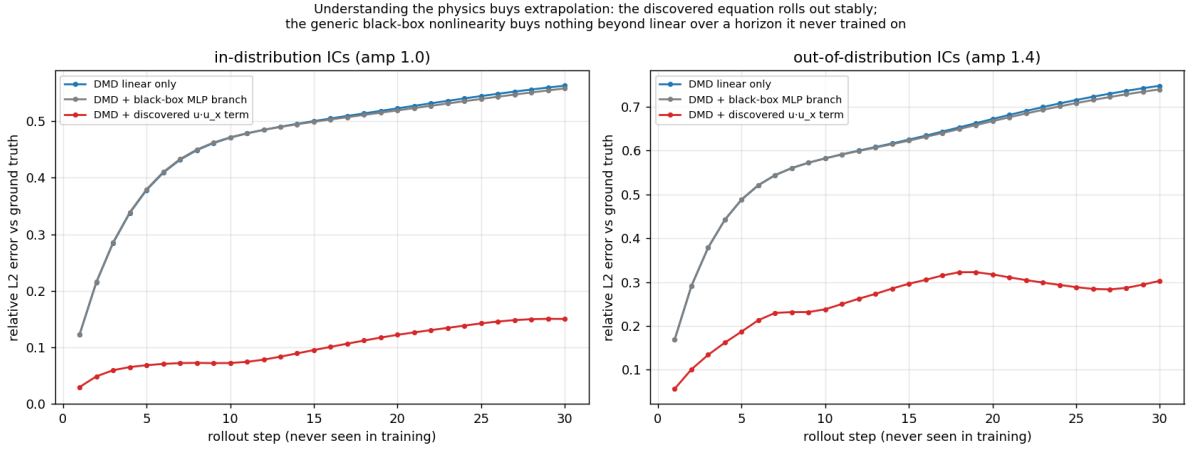


Figure 6: Extrapolation. Rollout error against step, in-distribution (left) and at unseen amplitude (right).

rollout training out-of-distribution (1.6%) — so it confirms stability but does *not* discriminate, and we report it as a check that passed rather than as evidence.

10 Related work

Learning solution operators for PDEs is an active field; Fourier neural operators [1] are, in the language of this paper, the fixed-spectral branch promoted to an operator. Koopman theory and DMD [2, 3] are the learned-linear branch, and the reduced-order-modelling literature has long used them for prediction. Sparse identification of nonlinear dynamics [4] and PDE-FIND [5] provide the reading step of Section 7; we use them unmodified. Recovering a coefficient field from spectral or boundary data is the classical inverse-conductivity problem [6], a mature area; our Section 4 is a small, well-specified parametric instance of it and claims no advance there. On the learned-representation side, it has been proved that invariant networks recover the Fourier transform of the underlying group and that the group is readable from the weights [7]; that result is stronger than anything we show about *why* classical bases appear, and we position our contribution as complementary and empirical. What we believe is not covered by the above is the packaging: a small set of *calibrated meters* with a shared shape — fixed structure plus measured failure — validated across two different objectives (a label-compression gate with

model	in-distribution	unseen amplitude
black-box branch, one-step trained	0.558 ± 0.000	0.739 ± 0.000
black-box branch, + rollout training	0.587 ± 0.036	0.814 ± 0.128
discovered uu_x , one-step trained	0.138 ± 0.012	0.249 ± 0.075
discovered uu_x, + rollout training	0.042 ± 0.003	0.074 ± 0.007

Table 4: Rollout error at step 30, three seeds. Rollout training improves the structurally correct model $\sim 3.4\times$ and makes the black box worse.

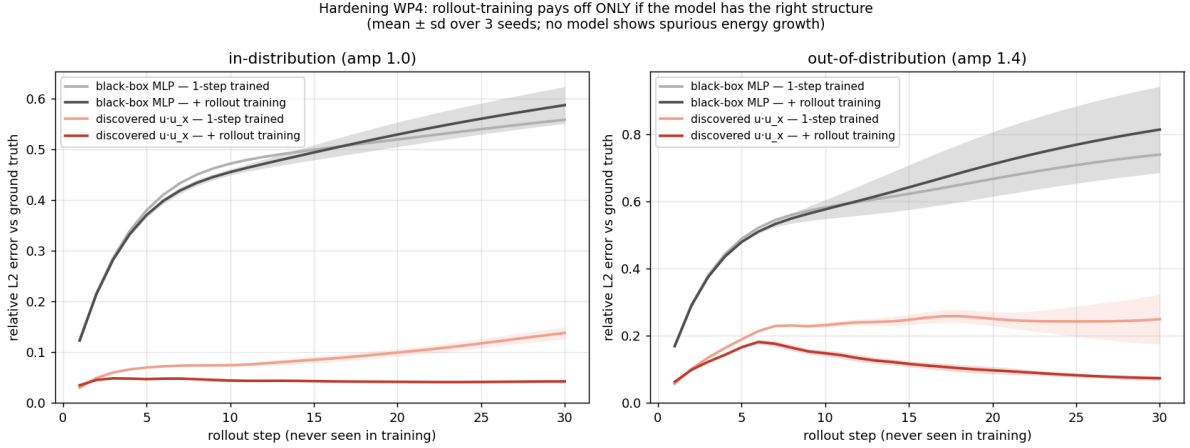


Figure 7: Hardening, three seeds (shaded: ± 1 s.d.). Rollout training pays off only for the model with the right structure.

$r = 0.96$, and the physical meters here), together with the controls that keep them honest.

11 Limitations and negative results

We state these explicitly, in the order they cost us time.

An artefact we had to remove. The heterogeneity meter was first built from a *prediction* error with Fourier-smooth initial conditions. It produced a *negative* gap — Fourier apparently better than the eigenbasis — because those initial conditions let Fourier compress the data irrespective of the dynamics, and a full linear head made the basis irrelevant. Moving to the operator-level metric of Equation (1) removed the confound.

An unfair comparison we had to fix. The first gated experiment trained a coupled network by stochastic gradient descent against a closed-form least-squares DMD and unsurprisingly lost, with a gate that did not behave as a meter. Freezing the DMD operator and training *only* the gated branch on its residual is what turned the gate into an instrument.

A training recipe that diverges. Rollout training from a cold network diverges (we observed exactly 1.000 relative error, i.e. a collapsed prediction). It must be applied as fine-tuning on a converged one-step model, with gradient clipping.

Seed sensitivity. Out-of-distribution rollout numbers vary substantially across seeds (± 0.075 at one-step); single-seed out-of-distribution claims should not be trusted, which is why Section 9 reports error bars and Section 8 should be read through them.

Scope. All experiments are one-dimensional, at $N = 64$, with a single nonlinear term whose form we ourselves planted. Recovery correlations of 1.0000 indicate a well-specified problem, not a difficult one: they validate the instrument, they do not demonstrate discovery on data

whose answer is unknown. Strong shocks are under-resolved. The linear head is a fitted DMD operator, not the exact diffusion operator, which caps how far even the structured model transfers in amplitude. Structure preservation is verified a posteriori (energy monitoring) rather than imposed by construction; a symplectic or conservative parametrization is the natural next hardening step.

12 Conclusion

Inverting the objective from classification to physical modelling makes the learned branch legible. In that regime the failure of a fixed spectral basis is a heterogeneity meter; the residual of a linear propagator is a nonlinearity meter; a gated learned branch fires with the same rectified law as the classification gate; the term that branch must contain is the term sparse regression names; and putting that named term into the model is what buys extrapolation — in horizon, in amplitude, and under rollout training, where structure is multiplied rather than replaced. The practical reading for a modeller is short: measure first whether your fixed front-end is failing and in which way, prefer a *structured* learned family over a free one, and treat the objective value itself as a mis-specification alarm.

13 Reproducibility

NumPy/SciPy/PyTorch with fixed seeds. Every experiment is a short deterministic script in `experiments/` that regenerates the figure of the same name — one per section, namely `phys_wp1_pilot`, `phys_wp1_close`, `phys_wp2_dmd`, `depth_test`, `phys_wp3_sindy`, `phys_wp4_rollout` and `phys_wp4b_hardening` — each accompanied by a `RESULTS` note stating its findings and its limits. Sparse regression is implemented directly (sequential thresholded least squares) with no external dependency. Trajectory generation for the Burgers sweep is cached in a regenerable archive; the scripts recompute it deterministically when the cache is absent.

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